

Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics

A Support-Visible OPH Route to the Clay Problem

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Draft of May 16, 2026

Paper release: r1514 **Released:** July 5, 2026

Abstract

The Clay Yang-Mills problem asks for a nontrivial four-dimensional quantum Yang-Mills theory with a positive mass gap. OPH gives a route on its compact-gauge branch.

The construction isolates the finite OPH repair mechanism and the certificate needed before that mechanism may be read as a four-dimensional Yang–Mills theorem. Fixed-cutoff local repair has a uniform rate floor and a positive transfer form. The Clay-facing statement is conditional on a renormalized continuum certificate: Schwinger-function convergence, reflection positivity, Euclidean covariance and locality, nontriviality, and Mosco or exact direct-limit convergence of the transfer/intertwiner data. Without that certificate, the finite repair branch is not by itself a construction of the four-dimensional Yang–Mills measure.

What This Paper Contributes

The standard target is clear: construct a nontrivial four-dimensional quantum Yang–Mills theory for every compact simple gauge group and prove a positive mass gap. OPH imports the usual compact gauge language, reflection positivity, and Osterwalder–Schrader/Wightman target conditions. The new ingredient is the repair interpretation of the vacuum sector.

On the support-visible compact-gauge branch, local holonomy data and the OPH scaling chart give the candidate Euclidean Yang–Mills form. Exact local repair acts as a positive Euclidean relaxation generator at finite cutoff. A uniform fixed-cutoff repair rate floor transports to the continuum Hamiltonian only when the transfer forms, vacuum projections, and finite intertwiners converge in the sense stated below. The proof therefore separates the finite repair mechanism from the additional four-dimensional continuum certificate required by the Clay problem.

1 Claim Boundary

This paper proves the finite positive-transfer and repair-gap mechanism on the sharpened support-visible exact Euclidean-consensus branch of OPH. The Clay-facing Yang–Mills theorem is conditional on the additional continuum certificate stated in Assumption [9.2](#). The branch data are

compact-gauge reconstruction, four-dimensional Euclidean scaling, reflection positivity, the ordinary zero-obstruction vacuum sector, exact local repair as conditional expectation, bounded-color active collar covers, repair completeness, and support-visible compact-gauge continuum extraction with transfer convergence.

More precisely, the imported four-dimensional form theorem uses compact-gauge reconstruction, the four-dimensional scaling chart, the reflection-positive ordinary vacuum sector, the absence of additional gauge-invariant relevant dimension-four pure-gauge operators beyond the positive quadratic curvature invariant, and the support-visible cylinder extraction. The later mass-gap step adds exact local repair, bounded-color active collars, and repair completeness.

Acceptance as a Clay-admissible solution depends on supplying the OPH compact-gauge continuum certificate: support-visible extraction, renormalized Schwinger-function convergence, reflection positivity, Euclidean covariance/locality, nontriviality, and transfer/intertwiner convergence at the strength required by the Clay/Jaffe–Witten statement. The proof below then isolates the finite repair mechanism and shows that, on any branch carrying that certificate, the gap is identified exactly:

$$\Delta_{\text{YM}} = \Delta_{\text{rep}}.$$

The two-dimensional heat-kernel identity in the wider OPH stack remains a separate normalization and worldsheet-effective bridge; it is not the mass-gap proof.

2 Position in the OPH Paper Stack

This paper is a focused companion to the OPH paper stack. The broad reconstruction program is summarized in *Observers Are All You Need* [1]. The compact technical core is *Recovering Relativity and the Standard Model from Observer Overlap Consistency* [2], which carries the support-visible compact-gauge repair-gap theorem inside the compact paper itself. The particle branch is separate [3]. The finite repair and quotient-normal-form machinery comes from *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [4]. The regulated screen, record, and edge heat-kernel architecture is developed in *Federated Echoshedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH* [5].

The edge-sector theorem in the stack relates OPH heat-kernel weights to a two-dimensional Yang–Mills partition identity. That identity fixes normalization and the controlled worldsheet effective bridge. The four-dimensional mass-gap argument uses repair dynamics: exact local repair becomes a positive Euclidean relaxation generator, and the uniform repair gap is transported to the compact-gauge Hamiltonian only under the continuum certificate.

The Clay-facing import is the compact paper’s D7 Yang–Mills theorem surface: the compact-gauge reconstruction ladder, the four-dimensional Euclidean Yang–Mills form theorem, the coherent compact-gauge extraction proposition, the conditional support-visible Osterwalder–Schrader reconstruction theorem, the nontriviality certificate, and the exact finite repair-gap theorem. This note isolates that branch theorem surface; it does not enlarge it.

3 The Clay Target

The Clay Mathematics Institute describes the Yang–Mills mass gap as the missing mathematical foundation behind the quantum theory used for nonabelian gauge forces [7]. Jaffe and Witten state the problem as follows: for every compact simple gauge group G , construct a nontrivial quantum Yang–Mills theory on $\mathbb{R}^4$ satisfying axiomatic properties at least as strong as the stated Wightman or Osterwalder–Schrader references and prove a mass gap $\Delta > 0$ [8, 9].

The OPH proof below addresses that target through a different variable. The starting data are finite observer patches, compact-gauge visible quotient data, and exact repair collars. The nonzero energy threshold is the cost of leaving the repair-fixed vacuum sector.

4 Standing Setup

Fix a compact simple gauge group G carried by an OPH compact-gauge zero-obstruction vacuum branch, realized at fixed cutoff by the declared compact-gauge patch-carrier architecture. The architecture supplies finite local Hilbert spaces, exact local constraints, patch and overlap algebras, overlap sector projectors, record layers, and local repair interfaces.

Let r range over a cofinal refinement family of finite regulators. For each r , let

$$(\mathcal{H}_r, \Omega_r, H_r), \quad T_r(t) = e^{-tH_r},$$

be the physical Euclidean Hilbert space, vacuum, Hamiltonian, and transfer semigroup.

Let X_r be the support-visible compact-gauge quotient state space, let π_r be the vacuum stationary measure, and set

$$K_r := L^2(X_r, \pi_r).$$

Let $\mathcal{C}_r$ be the finite family of active repair collars. For each active collar $C \in \mathcal{C}_r$, let

$$\rho_C : X_r \rightarrow Y_C$$

be the complete repaired visible datum, and let

$$E_C : K_r \rightarrow K_r$$

be conditional expectation onto the ρ_C -measurable functions.

The proof stays on the ordinary or central zero-obstruction vacuum branch. The genuinely noncentral higher-gauge branch is a different fixed-cutoff sector in the OPH stack and is not used for the ordinary compact-simple G theorem below.

Notation	Meaning
X_r, π_r, K_r	support-visible quotient state space, vacuum stationary measure, and $L^2(X_r, \pi_r)$.
$\mathcal{C}_r$	finite active repair-collar family at regulator r .
ρ_C	complete repaired visible datum on collar C .

Notation	Meaning
E_C	π_r -preserving conditional expectation onto ρ_C -measurable functions.
L_r^{rep}	ground-state transformed Euclidean repair generator at cutoff r .
$P_{0,r}$	projection onto constants in K_r , corresponding to the physical vacuum.
U_r	finite-stage unitary from the physical Euclidean Hilbert space to the repair L^2 space.
$K, \mathcal{H}, U$	support-visible continuum repair Hilbert space, physical Hilbert space, and limiting unitary.
c_*	uniform active-collar repair-rate floor.

5 Imported 4D Euclidean Yang–Mills Form

Assumption 5.1 (Support-visible compact-gauge Yang–Mills branch). *The compact-gauge branch used below satisfies the following branch-local conditions.*

- (i) *the ordinary or central zero-obstruction compact-gauge sector survives refinement with compact simple structure group G ;*
- (ii) *the support-visible quotient carries a four-dimensional Euclidean scaling chart;*
- (iii) *the ordinary vacuum sector is reflection positive and has topological angle $\theta = 0$;*
- (iv) *the local finite-constraint MaxEnt/Gibbs family is gauge-invariant, Euclidean local, rotation-invariant, and refinement-stable;*
- (v) *no additional gauge-invariant relevant dimension-four pure-gauge operator remains on this branch besides the positive quadratic curvature invariant;*
- (vi) *the support-visible compact-gauge cylinder family has the weak-* / GNS continuum extraction stated in Proposition 9.1.*

Theorem 5.2 (Imported OPH branch theorem: four-dimensional Euclidean Yang–Mills form). *Under Assumption 5.1, the continuum gauge-sector Euclidean action is*

$$S_E[A] = \frac{1}{4g^2} \int_{\mathbb{R}^4} \langle F_{\mu\nu}, F_{\mu\nu} \rangle d^4x, \quad F = dA + A \wedge A, \quad (\text{YM})$$

with compact simple structure group G . Equivalently, the support-visible continuum transfer semigroup is the Euclidean Yang–Mills semigroup associated with the gauge-quotient cylinder measure

$$d\mu_{\text{YM}}(A) = Z^{-1} e^{-S_E[A]} DA/G$$

in the OPH support-visible GNS representation.

Proof. This is the support-visible compact-gauge Yang–Mills form theorem of the compact OPH paper, where it appears as Theorem `thm:oph-4d-euclidean-yang-mills-form` [2]. The proof spine is recalled here because it fixes the target Hamiltonian for the mass-gap step.

The compact-gauge branch reconstructs a compact group G from the zero-obstruction transportable bosonic sector category and its faithful fiber functor [2]. At fixed cutoff, the declared compact-gauge patch-carrier presentation gives support-visible link holonomies and plaquette holonomies. In the refinement limit, the zero-obstruction gluing law makes infinitesimal rectangle holonomies multiplicative and path-local. Therefore there is a local connection A on the four-dimensional scaling chart, and the infinitesimal plaquette defect is

$$U_{\mu\nu}(\varepsilon, x) = \mathbf{1} + \varepsilon^2 F_{\mu\nu}(x) + O(\varepsilon^3), \quad F = dA + A \wedge A.$$

The Euclideanized MaxEnt/local-Gibbs branch gives a local finite-range action density built from support-visible gauge-invariant collar data. Gauge quotienting permits only class functions of the curvature and its covariant derivatives. The four-dimensional scaling chart, Euclidean rotation invariance, locality, and reflection positivity leave one relevant dimension-four positive quadratic invariant in the pure gauge sector:

$$\langle F_{\mu\nu}, F_{\mu\nu} \rangle.$$

The possible topological density $\langle F \wedge F \rangle$ is reflection odd and belongs to a separate topological-angle sector; it is absent on the ordinary reflection-positive zero-obstruction vacuum branch used here. Higher curvature powers and covariant-derivative terms are irrelevant operators under the declared continuum scaling and vanish from the strict Yang–Mills fixed form. Normalizing the unique positive quadratic invariant defines the coupling g and gives (YM).

The finite-stage cylinder measures are the gauge-register / quantum-link Gibbs measures pushed to the support-visible quotient. Assumption 5.1 supplies the compatible support-visible weak-* / GNS cylinder extraction. The resulting Euclidean transfer semigroup is therefore the transfer semigroup of (YM). $\square$

Corollary 5.3 (Yang–Mills equations on the compact-gauge branch). *With an external conserved gauge current J , the OPH curvature-square action on the same support-visible compact-gauge branch gives*

$$DF = 0, \quad D*F = g^2 *J,$$

equivalently

$$D_\mu F^{\mu\nu} = g^2 J^\nu$$

in local coordinates after the usual continuation to the Lorentzian field-equation convention.

Proof. The curvature $F = dA + A \wedge A$ obeys the covariant Bianchi identity $DF = 0$. Varying the quadratic curvature action

$$S[A, J] = -\frac{1}{2g^2} \int \langle F \wedge *F \rangle + \int \langle A \wedge *J \rangle$$

with respect to compactly supported variations of A gives

$$\delta S = \int \langle \delta A \wedge (g^{-2} D*F - *J) \rangle$$

up to a boundary term. Stationarity gives $D*F = g^2 *J$. The source-free equation is the case $J = 0$. $\square$

Remark 5.4 (Why this step matters for the prize). The proof has two separate claims. Theorem 5.2 identifies the support-visible continuum gauge sector with four-dimensional Euclidean Yang–Mills under its branch hypotheses. The spectral argument applies to that Hamiltonian and proves a positive gap only when Assumption 9.2 supplies the OS/transfer/nontriviality bridge.

Remark 5.5 (Abelian boundary and Maxwell branch). The ordinary electromagnetic $U(1)_Q$ branch is the abelian boundary case of the same compact-gauge curvature package. Restricting $F = dA + A \wedge A$ to $\mathfrak{u}(1)_Q$ gives

$$F_Q = dA_Q, \quad dF_Q = 0, \quad d*F_Q = g_Q^2 *J_Q$$

after varying the quadratic electromagnetic action with current J_Q . This recovers Maxwell’s equations after canonical electromagnetic normalization. The Clay-facing mass-gap claim concerns compact simple nonabelian G on the support-visible branch above. The abelian photon branch is gapless because the unbroken $U(1)_Q$ gauge symmetry protects the massless carrier.

6 Fixed-Cutoff Repair Equals Projection

Proposition 6.1 (Local exact repair equals conditional expectation). *For each active collar C , the exact-Markov repair map on the support-visible quotient is the π_r -preserving conditional expectation E_C .*

Proof. On the exact-Markov branch, repair preserves exactly the repaired visible datum ρ_C , changes only complementary invisible fiber data, and acts on the quotient-first physical algebra rather than on microscopic representatives. Let Φ_C be the Heisenberg repair map and let $\mathcal{N}_C$ be the repaired local fixed algebra. The repair semantics give

$$\Phi_C(a) = a \quad (a \in \mathcal{N}_C),$$

$$\Phi_C(\mathcal{A}_r^{\text{sv}}) \subseteq \mathcal{N}_C, \quad \pi_r \circ \Phi_C = \pi_r,$$

and Φ_C is $\mathcal{N}_C$ -bimodular. Therefore, for $a \in \mathcal{N}_C$ and $x \in \mathcal{A}_r^{\text{sv}}$,

$$\pi_r(a^* \Phi_C(x)) = \pi_r(a^* x).$$

Since $\Phi_C(x) \in \mathcal{N}_C$ and π_r is faithful, this equation uniquely characterizes $\Phi_C(x)$ as the orthogonal projection of x onto $\mathcal{N}_C$ in the GNS inner product. That orthogonal projection is the π_r -preserving conditional expectation E_C . $\square$

7 Exact Euclidean-Consensus Law

Lemma 7.1 (Fiber-homogeneous orbit condition). *Fix an active collar C and a repaired value $y \in Y_C$. If, on the support-visible quotient, the hidden fiber*

$$F_C(y) = \rho_C^{-1}(y)$$

has no remaining observable labels, the conditioned local MaxEnt state is uniform on $F_C(y)$, and the primitive collar relaxation is invariant under all measure-preserving permutations of $F_C(y)$, then the fiber is homogeneous for the local repair receipt.

Proof. This is the finite fiber-homogeneity receipt. Quotienting removes declared implementation labels, but the proof must also check that no remaining source constraint or relaxation rate distinguishes points in the fiber. Under that check, the conditioned MaxEnt state and primitive relaxation carry the stated permutation symmetry. $\square$

Lemma 7.2 (Scalar relaxation on a uniform hidden fiber). *Let F be a finite hidden fiber with uniform measure, let E_F be expectation onto constants, and let D_F be a positive self-adjoint Markov relaxation generator such that*

$$\ker D_F = \text{Ran}(E_F)$$

and D_F commutes with the full permutation group of F . Then there is a scalar $c_F > 0$ such that

$$D_F = c_F(I - E_F).$$

Proof. The permutation representation on $L^2(F)$ splits as constants plus the zero-sum subspace. The zero-sum subspace is irreducible for the full symmetric group when $|F| \geq 2$. Schur's lemma therefore makes D_F a scalar on that subspace. Positivity and the kernel condition make the scalar strictly positive. The formula follows. $\square$

Theorem 7.3 (Exact Euclidean repair law under homogeneous fibers). *Under the fiber-homogeneous orbit condition for every active collar, there are positive constants $c_C > 0$ such that the ground-state transformed physical Euclidean generator is exactly*

$$L_r^{\text{rep}} = \sum_{C \in \mathcal{C}_r} c_C(I - E_C), \quad (1)$$

and therefore

$$U_r e^{-tH_r} U_r^{-1} = e^{-tL_r^{\text{rep}}} \quad (t \geq 0), \quad (2)$$

for a unitary $U_r : \mathcal{H}_r \rightarrow K_r$ with $U_r \Omega_r = \mathbf{1}_r$.

Proof. The OPH MaxEnt axiom gives a local-Gibbs state with a quasi-local finite-range generator on the declared finite-constraint branch. After the support-visible quotient, each primitive local Euclidean piece D_C is supported on one collar C , preserves exactly the repaired visible datum ρ_C , and relaxes only complementary fiber data. Hence

$$\ker D_C = \text{Ran}(E_C).$$

Lemma 7.1 gives the full hidden-fiber permutation symmetry. Applying Lemma 7.2 fiberwise gives a positive scalar c_C on the orthogonal complement of the repaired datum:

$$D_C = c_C(I - E_C).$$

The scalar is positive because C is active. Branch homogeneity makes it a collar-type scalar, so summing over the active collars gives (1), and exponentiation of the positive self-adjoint generator gives (2). $\square$

Lemma 7.4 (Uniform active-collar rate floor). *Assume finite local combinatorial type, a branch-homogeneous local constraint family, and refinement-stable exact repair semantics. Then the local Euclidean repair rate depends only on active collar type, the active collar-type set is finite across the cofinal refinement family, and there is $c_* > 0$, independent of r , such that*

$$c_C \geq c_* \quad \text{for every active collar } C \in \mathcal{C}_r. \quad (3)$$

Proof. Active collars are exactly the collars on which complementary invisible fiber data are genuinely relaxed. Hence $D_C \neq 0$ on $\ker(E_C)$, so $c_C > 0$. Finite local combinatorial type gives a finite list of bounded collar patterns, including their visible interface alphabets, hidden fiber cardinalities, local constraint templates, and admissible repair maps. Branch homogeneity makes the conditioned MaxEnt relaxation scalar a function of this finite type data. Refinement stability says that refinement replaces a collar by copies of the same bounded type templates with the same normalized local Euclidean repair semantics, so no new rate-degenerating collar type appears along the cofinal family. Taking the minimum of c_C over the finite active type list gives $c_* > 0$. $\square$

8 Finite-Stage Gap

Proposition 8.1 (Special commuting-color finite-stage repair gap). *Assume that the collar family admits a bounded-color decomposition*

$$\mathcal{C}_r = \bigsqcup_{a=1}^q \mathcal{C}_{r,a}, \quad q < \infty, \quad (4)$$

independent of r , and that exact quotient-local gluing makes the corresponding color expectations commute. Then

$$L_r^{\text{rep}} \geq c_*(I - P_{0,r}), \quad (5)$$

where $P_{0,r}$ projects onto constants in K_r . Hence

$$\Delta_{\text{rep},r} \geq c_* > 0. \quad (6)$$

Proof. For each color a , define the parallel color expectation

$$E_{r,a} := \prod_{C \in \mathcal{C}_{r,a}} E_C.$$

Same-color collars are disjoint, so the factors commute. Exact quotient-local gluing makes $\{E_{r,a}\}_{a=1}^q$ a commuting family of orthogonal projections on the vacuum branch.

Repair completeness says that the only support-visible observables fixed by every local repair are constants:

$$\bigcap_{a=1}^q \text{Ran}(E_{r,a}) = \mathbb{C} \mathbf{1}_r. \quad (7)$$

Define

$$\tilde{L}_r := \sum_{a=1}^q c_*(I - E_{r,a}).$$

For commuting projections,

$$I - \prod_C E_C \leq \sum_C (I - E_C),$$

so $\tilde{L}_r \leq L_r^{\text{rep}}$. The projections $E_{r,a}$ are simultaneously diagonalizable. On the joint all-ones eigenspace, (7) says the vector is constant. Every nonconstant joint eigenspace has at least one color eigenvalue zero, so

$$\tilde{L}_r \geq c_*(I - P_{0,r}).$$

This proves (5), and the spectral gap bound (6) follows. $\square$

Remark 8.2 (Generic repair-gap certificate). The commuting-color proposition is a special certificate, not the generic repair-generator route. For overlapping collars one must instead compute

$$\kappa_r = \inf_{f \perp 1} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If local fiber rates have a lower bound $\gamma_* > 0$, then

$$L_r^{\text{rep}} \geq \gamma_* \kappa_r (I - P_{0,r}).$$

Finite repair completeness gives $\kappa_r > 0$ at fixed regulator; a uniform refinement lower bound is a separate receipt.

9 Continuum Extraction

Proposition 9.1 (Coherent refinement and support-visible extraction). *Under Assumption 5.1, the compact-gauge branch supplies the coherence and extraction data needed for the continuum intertwining theorem:*

- (1) *finite-stage support-visible quotient state spaces X_r , algebras $\mathcal{A}_r^{\text{sv}}$, stationary states π_r , and Hilbert spaces $K_r = L^2(X_r, \pi_r)$;*
- (2) *refinement maps $j_{rs}^{\text{rep}} : K_r \rightarrow K_s$ and $j_{rs}^{\text{phys}} : \mathcal{H}_r \rightarrow \mathcal{H}_s$ compatible with all repaired local marginals;*
- (3) *a projectively compatible family of finite-stage compact-gauge cylinder marginals;*
- (4) *weak-* compactness and a diagonal subnet whose local marginals converge on every support-visible cylinder algebra;*
- (5) *support-visible GNS gluing to continuum Hilbert spaces K and $\mathcal{H}$, with dense images of the local cylinder algebras;*
- (6) *a faithful limiting vacuum pair on the support-visible gauge-invariant algebra after quotienting the maximal repair-invariant overlap-trivial kernel.*

Proof. The compact-gauge ladder in the OPH compact paper proves fixed-cutoff bosonic sector categories, faithful monoidal refinement functors, compatible finite-dimensional fibers, the directed colimit sector category, and compact gauge reconstruction [2]. The support-visible compact-gauge quotient algebra is generated by overlap sector projectors and compact-gauge visible bosonic observables, modulo the maximal repair-invariant overlap-trivial kernel.

Each regulator has finite-dimensional state space and finite local cylinder algebras, so its state space is weak-* compact. Refinement compatibility makes the local marginals projective: the restriction of a refined cylinder marginal to any coarser cylinder equals the coarser marginal. Choosing a countable cofinal family of support-visible cylinders and applying a diagonal subnet argument gives a limiting state on their algebraic union. Positivity and normalization pass to the limit on every cylinder, and the quotient by the repair-invariant overlap-trivial kernel makes the limit faithful on the support-visible gauge-invariant algebra.

The GNS construction applied to this limiting state gives K . Transporting the same coherent refinement data through the finite-stage intertwiners U_r gives the physical limiting Hilbert space $\mathcal{H}$ only as a candidate support-visible Hilbert object. Weak- $*$ convergence of cylinder states does not by itself prove convergence of transfer generators, vacuum projections, or intertwiners. $\square$

Assumption 9.2 (Four-dimensional Yang–Mills continuum certificate). *In addition to the finite compact-gauge repair data, one regulator family $r = (a_s, a_t, L, b)$ supplies all of the following.*

- (Y1) *Renormalized gauge-invariant cylinder maps $\mathcal{R}_r : \mathfrak{A}_0^G(K) \rightarrow \mathcal{A}_r^G(K_r)$ and a uniform finite-volume/lattice-spacing Cauchy bound on every bounded local test family.*
- (Y2) *Exact finite reflection positivity, or a positive self-adjoint transfer matrix, with reflection-compatible renormalization maps.*
- (Y3) *Uniform restoration of Euclidean covariance, locality, regularity, clustering, and the declared vacuum-sector condition.*
- (Y4) *Generalized Mosco convergence of the physical transfer forms q_r to a closed nonnegative continuum form q , or exact coherent direct-limit semigroup identities.*
- (Y5) *Convergence of the finite intertwiners U_r and their inverses on dense form cores to a unitary U .*
- (Y6) *A nontriviality certificate: renormalized local observables A_r with*

$$\inf_r \text{Var}_{\omega_r}(A_r) > 0, \quad \sup_r q_r(A_r \Omega_r) < \infty.$$

The explicit multiresolution construction in the main paper supplies a model for items (Y1)–(Y2) and an exact transfer tower when its shell identities are implemented. Identification of its continuum limit with the four-dimensional Yang–Mills OS theory is the additional content of (Y3)–(Y6).

Theorem 9.3 (Conditional continuum transfer identification). *Under Assumption 9.2, the generalized Mosco limit produces nonnegative self-adjoint generators H and L^{rep} and a unitary U satisfying*

$$U e^{-tH} U^{-1} = e^{-tL^{\text{rep}}} \quad (t \geq 0), \quad (8)$$

and hence

$$U H U^{-1} = L^{\text{rep}}. \quad (9)$$

If the finite repair generators satisfy

$$L_r^{\text{rep}} \geq c_*(I - P_{0,r})$$

with one $c_ > 0$ and the vacuum projections converge strongly, then*

$$H \geq c_*(I - P_0).$$

Proof. Generalized Mosco convergence is equivalent to generalized strong-resolvent and transfer-semigroup convergence on the declared Hilbert identifications. Convergence of U_r and U_r^{-1} on dense form cores passes the finite-stage intertwining relation to the limit, so uniqueness of self-adjoint semigroup generators gives $U H U^{-1} = L^{\text{rep}}$. For a Mosco recovery sequence $\psi_r \rightarrow \psi$,

$$q_r(\psi_r) \geq c_*(\|\psi_r\|^2 - \|P_{0,r}\psi_r\|^2).$$

Taking the limit and using strong convergence of the vacuum projections gives the continuum form inequality. $\square$

10 Axiomatic Reconstruction and Nontriviality

Theorem 10.1 (Conditional Osterwalder–Schrader reconstruction on the support-visible compact-gauge branch). *Under Assumptions 5.1 and 9.2, the continuum support-visible compact-gauge cylinder family is Euclidean invariant, reflection positive, regular on gauge-invariant local cylinder observables, nontrivial, and cyclic for the vacuum sector. Hence Osterwalder–Schrader reconstruction gives a four-dimensional quantum Yang–Mills theory*

$$(\mathcal{H}, \Omega, H, \mathcal{A}_{\text{loc}}^G)$$

on the support-visible gauge-invariant local algebra, with $H \geq 0$ and e^{-tH} equal to the certified continuum Euclidean transfer semigroup.

Proof. Euclidean covariance, locality, reflection positivity, regularity, clustering, transfer convergence, and nontriviality are precisely the certificate items in Assumption 9.2. Proposition 9.1 supplies the support-visible cylinder algebra and candidate GNS objects; the certificate supplies the missing continuum and transfer controls. The standard Osterwalder–Schrader reconstruction theorem therefore produces the Hilbert space, vacuum, local algebra, and positive Hamiltonian, and identifies the physical time-translation semigroup with the certified Euclidean transfer semigroup [10, 11]. $\square$

Proposition 10.2 (Nontriviality of the support-visible compact-gauge theory). *The support-visible compact-gauge local algebra on the zero-obstruction vacuum branch strictly contains the vacuum scalars and admits a non-vacuum finite-energy local excitation.*

Proof. The compact-gauge witness and physical-UV landing theorem in the OPH compact paper supplies a realized nontrivial compact-gauge branch [2]. At finite cutoff this gives a support-visible gauge-invariant local observable, such as a nonconstant Wilson/plaquette cylinder observable, whose vacuum variance is positive. Its GNS vector is orthogonal to the vacuum after subtracting its expectation value. The finite-range local-Gibbs generator assigns finite energy to finite-cylinder excitations. The continuum nontriviality step is not a consequence of weak-* state convergence alone; it is the variance-floor and finite-energy item (Y6) in Assumption 9.2. With that certificate, the local cylinder vectors survive in the support-visible continuum, so the continuum local algebra is strictly larger than $\mathbb{C}I$ and contains non-vacuum finite-energy local excitations. $\square$

11 Main Theorem

Theorem 11.1 (Conditional positive four-dimensional compact-gauge Yang–Mills mass gap). *Let G be a compact simple gauge group carried by a support-visible compact-gauge OPH vacuum branch satisfying the standing setup above and the continuum certificate of Assumption 9.2. The theory reconstructed in Theorem 10.1 is nontrivial, and its continuum support-visible Hamiltonian H satisfies*

$$H \geq c_*(I - P_0), \tag{10}$$

where P_0 projects onto the vacuum. Therefore

$$\text{Spec}(H) \cap (0, c_*) = \emptyset, \quad \Delta_{\text{YM}} \geq c_* > 0. \tag{11}$$

Moreover, on that certified continuum branch the repair gap and Yang–Mills gap are exactly equal:

$$\Delta_{\text{YM}} = \Delta_{\text{rep}}. \quad (12)$$

Proof. At every finite stage, Proposition 8.1 gives

$$L_r^{\text{rep}} \geq c_*(I - P_{0,r}).$$

By Proposition 9.1 and Theorem 9.3, this lower bound passes to the support-visible continuum:

$$L^{\text{rep}} \geq c_*(I - P_0).$$

Using (9), $UHU^{-1} = L^{\text{rep}}$. Conjugating the lower bound by U^{-1} gives (10), and the spectral statement (11) follows. Since unitary equivalence preserves the nonzero spectrum and identifies the vacuum with the constant sector, the infimum of the nonzero spectrum is the same on both sides, giving (12). $\square$

Remark 11.2 (Group-uniform form). The proof is group-uniform. Once a compact simple G is carried by a compact-gauge zero-obstruction OPH branch, no step uses special properties of the realized Standard Model quotient. The inputs are compact-gauge support-visible quotient locality, exact local repair on collars, bounded-color collar covers, refinement coherence, and support-visible continuum extraction.

12 Exact Gap Accounting

On a branch satisfying Assumption 9.2, the proof gives more than a positive lower bound. It identifies the Hamiltonian whose gap is being measured:

$$H = U^{-1}L^{\text{rep}}U.$$

Therefore

$$\text{Spec}(H) \setminus \{0\} = \text{Spec}(L^{\text{rep}}) \setminus \{0\}.$$

The Yang–Mills gap is exactly the first nonzero repair eigenvalue:

$$\Delta_{\text{YM}} := \inf(\text{Spec}(H) \setminus \{0\}) = \inf(\text{Spec}(L^{\text{rep}}) \setminus \{0\}) =: \Delta_{\text{rep}}.$$

The finite-stage color argument proves $\Delta_{\text{rep}} \geq c_* > 0$. The exact accounting statement is the conditional equality $\Delta_{\text{YM}} = \Delta_{\text{rep}}$; the inequality is the positivity proof for that same quantity. Without the continuum certificate, the same equations remain finite-regulator repair accounting instead of a Clay-admissible Yang–Mills theorem.

13 Relation to the 2D Yang–Mills Bridge

The OPH corpus also contains an exact edge-sector identity:

$$Z_{\text{edge}}(t) = \sum_R d_R^2 e^{-tC_2(R)} = K_t(1),$$

which is the compact-group heat kernel at the identity and matches the standard two-dimensional Yang–Mills heat-kernel partition form. Peter–Weyl supplies the heat-kernel identity, and Gross–Taylor gives the standard large- N worldsheet dictionary when a separate large- N_{edge} branch with remainder control is declared [12, 13].

That 2D result is a normalization and worldsheet-effective bridge. The spectral theorem above concerns the support-visible compact-gauge Hamiltonian and obtains its lower bound by identifying Euclidean transfer with the repair generator. The string-vacuum selector uses the same bridge only as effective edge-language input; its critical-worldsheet, Bouchard–Donagi, safety-layer, threshold, and moduli-locking gates do not alter the four-dimensional compact-gauge mass-gap theorem surface [6].

14 Clay Deliverables Checklist

Clay/Jaffe–Witten target	OPH repair-dynamics boundary
Compact simple gauge group G	G is arbitrary compact simple, provided it is carried by the OPH compact-gauge zero-obstruction branch.
Four-dimensional Euclidean Yang–Mills form	Theorem 5.2, imported from the compact paper’s D7 Yang–Mills theorem surface, gives $S_E[A] = \frac{1}{4g^2} \int_{\mathbb{R}^4} \langle F_{\mu\nu}, F_{\mu\nu} \rangle d^4x$ under the branch assumptions in Assumption 5.1.
Four-dimensional quantum Yang–Mills theory	Conditional on Assumption 9.2: support-visible extraction, renormalized Schwinger-function convergence, reflection positivity, Euclidean covariance/locality, clustering, nontriviality, and transfer/intertwiner convergence.
Nontriviality	Conditional on the variance-floor and finite-energy certificate in item (Y6). Finite nonconstant Wilson/plaquette cylinders are candidates; weak- $*$ convergence alone is not enough.
Axiomatic strength	Theorem 10.1 states the conditional OS reconstruction step on the support-visible gauge-invariant local algebra.
Mass gap	The finite repair generator satisfies $L_r^{\text{rep}} \geq c_*(I - P_{0,r})$. The continuum Hamiltonian bound $H \geq c_*(I - P_0)$ follows only after Theorem 9.3.
Exact gap accounting	On the certified continuum branch, $UHU^{-1} = L^{\text{rep}}$ gives $\Delta_{\text{YM}} = \Delta_{\text{rep}}$. OPH accounts exactly for the Yang–Mills gap as the first nonzero repair eigenvalue only on that branch.
Uniformity in G	The proof uses compact simplicity and compact-gauge quotient locality, not Standard-Model-specific representation data.

15 Conclusion

The OPH finite repair mechanism for the Yang–Mills mass gap is simple. A non-vacuum support-visible compact-gauge excitation is not fixed by all local repair collars. At least one active collar must relax it. Because active collar rates have a uniform positive floor and the collars can be organized into finitely many commuting colors, every non-vacuum finite-regulator state pays a positive Euclidean repair cost.

The Clay-facing conclusion is conditional. If the regulator family also satisfies the renormalized continuum, reflection-positivity, OS, nontriviality, and transfer/intertwiner certificate of Assumption 9.2, then the continuum compact-gauge Hamiltonian is unitarily equivalent to the repair generator. On that certified branch the Yang–Mills gap is the repair gap:

$$\Delta_{\text{YM}} = \Delta_{\text{rep}} \geq c_* > 0.$$

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